Whole-Body Control for a Mobile Manipulator with QP + Reinforcement Learning

RoArm-M3 demo: MTC planning and camera-guided grasping

Project overview

This poster presents the RoArm-M3 motion-planning and vision-guided grasping components of the wider whole-body control project. A fixed camera provides a target pose; eye-to-hand calibration maps it into the robot base frame. MoveIt Task Constructor (MTC) plans the manipulation stages, while measured joint feedback updates the robot state.

Demo pipeline

01 Camera — Target observation

→

02 Calibration — Camera → base

→

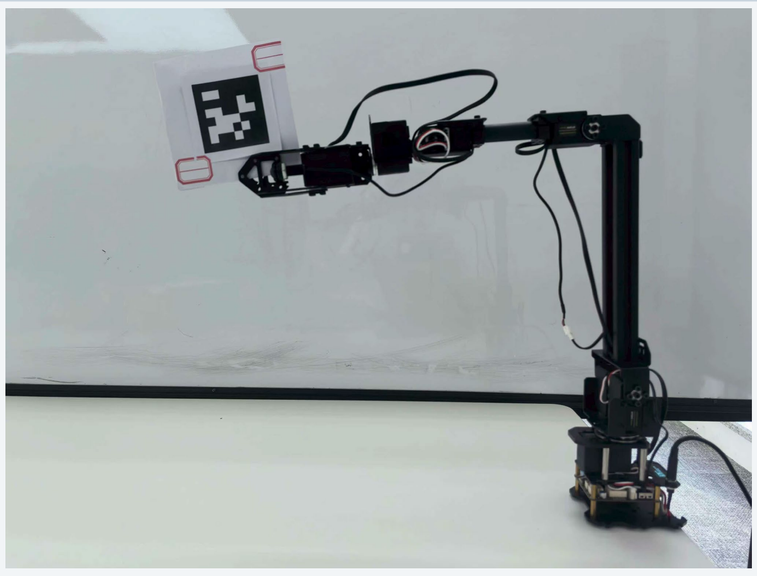
03 MTC — Approach + grasp

→

04 RoArm-M3 — Execute + feedback

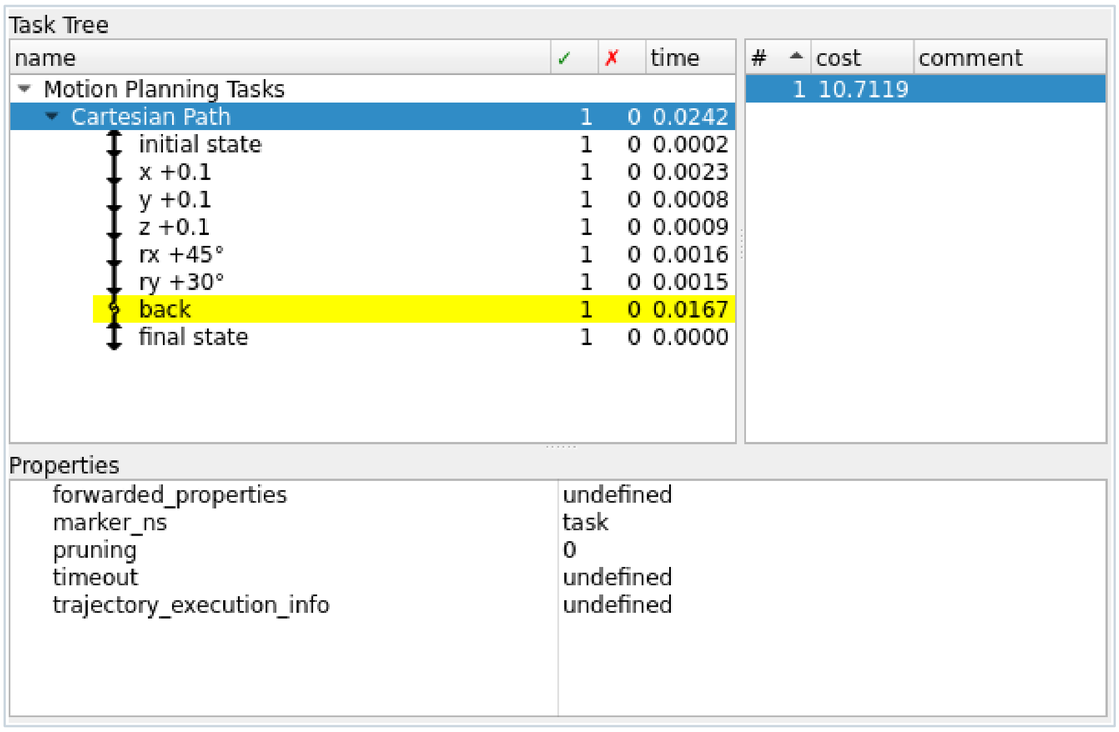
Control path: MoveIt 2 / MTC → ros2_control → serial driver → RoArm-M3. Measured joint angles → /joint_states → TF.

Existing hardware setup



RoArm-M3 with AprilTag at the gripper for eye-to-hand calibration.

MoveIt Task Constructor plan



Provided MTC Task Tree: one displayed Cartesian Path solution. This image does not show grasp execution.

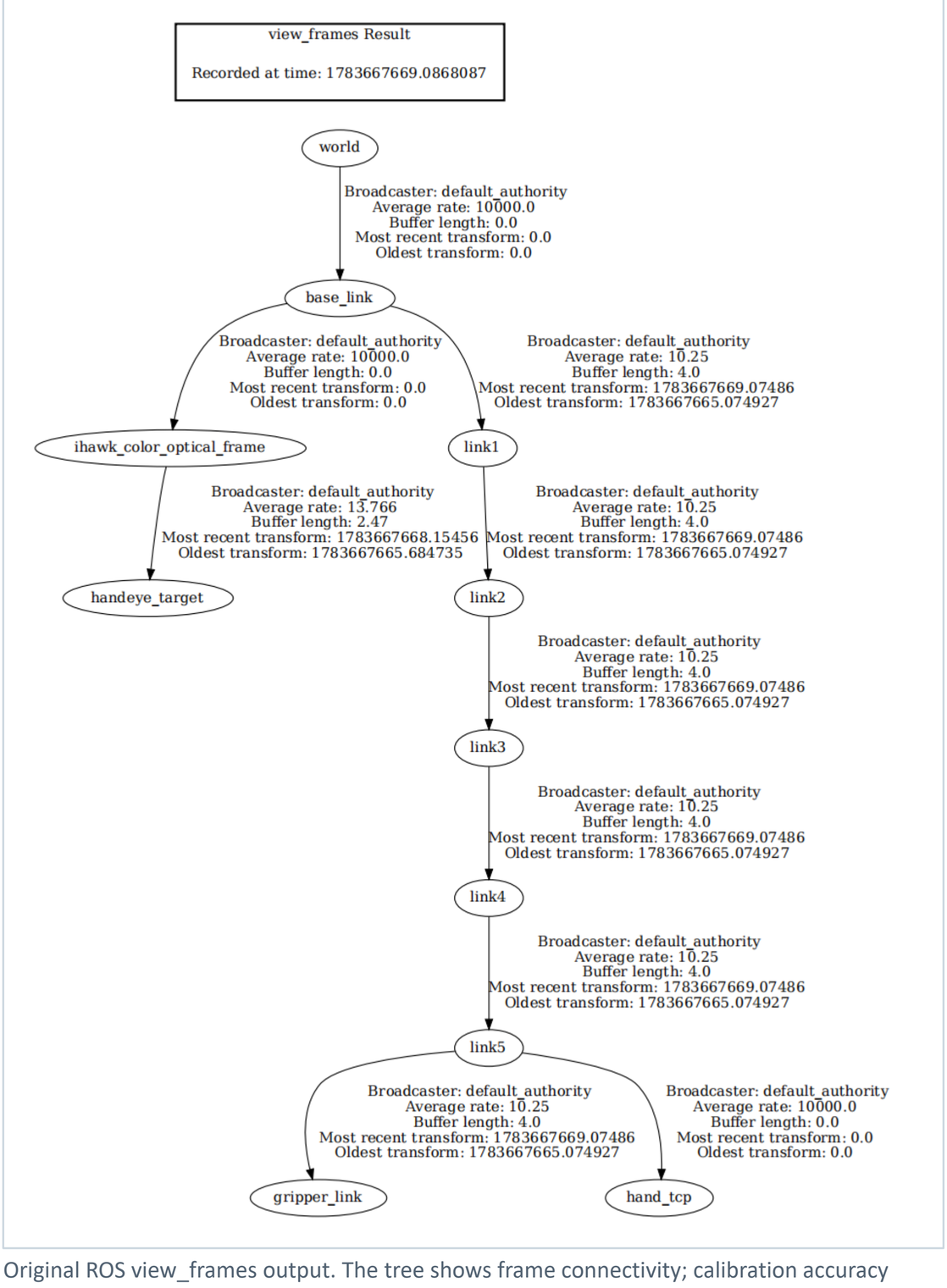
Grasp trials

17 / 20 successful trials · 85%

95% exact CI: 62.1–96.8% | an irregularly shaped part; randomized starts; real hardware

ESP32 LSTM controller; 30 demonstrations; one evaluated checkpoint.

Complete TF tree



Original ROS view_frames output. The tree shows frame connectivity; calibration accuracy needs separate validation.

Accuracy

End-effector position: 2.96 mm mean ± 0.57 mm SD (n=6); held-out split unspecified

Camera reprojection: 0.30 px mean (n=25); calibration-set fit

Camera-to-robot alignment: 1.18 mm RMS (n=12); correspondence fit

Trajectory tracking: 2.9 mm MAE; sample count not reported

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